A Surface-Content Decomposition of AI-Generated Text Detection Benchmarks: Evidence from Five Models on DAIGT V2 and HC3

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***Abstract*—A detector that scores 99% on a machine-generated-text benchmark may have modelled how machine language differs from human language, or how one corpus happened to be punctuated and encoded, and the headline number does not distinguish the two. We propose a decomposition that measures the difference. A surface-only model reads 47 orthographic features and never sees a word, a content-only model reads text with punctuation, casing and non-ASCII stripped, and a fine-tuned transformer is reported as a reference point. On HC3 the two arms are indistinguishable under a paired test, 3.20% against 3.26% error, and the null holds on five of five independent partitions with the sign changing between them, while DeBERTa reaches 0.28% on the same split, so the parity is between two weak arms rather than a ceiling. On DAIGT V2 content is stronger by a factor of 7.9 in error. Splitting each corpus by its own subgroup labels localises the HC3 result rather than confirming it. The surface arm reaches 0.01% error on the Reddit sub-corpus that is 74.8% of the whole and loses to content on both other domains large enough to support a claim, and removing the one cue responsible with the cleaning kit published alongside the corpus costs it 10.03 points. A tokenisation control then shows BERT emits identical token identifiers for strings differing only by that cue, so it cannot represent the cue and still reaches 0.9916 weighted F1, making the cue sufficient in isolation but unnecessary in practice. Surface form is informative on both corpora. We claim only that it reaches parity with content on one, and there for one reason.**

***Index Terms*—AI-generated text detection, benchmark evaluation, surface features, shortcut learning, tokenisation**

I. Introduction

Detectors of machine-generated text report accuracy at or above 97% on the benchmarks the field uses most, and frequently above 99% [1], [2], [3]. Read directly, those numbers say the task is close to solved. Read carefully, they say something weaker, because a benchmark score measures the separability of a particular corpus and not the capability a reader infers from it.

Separability has more than one source. A detector can reach a high score by modelling how machine-generated language differs from human language, or by modelling how one corpus was assembled, punctuated and encoded. Both produce the same headline figure. Individual instances of the second are documented, including a whitespace convention in HC3 [4], a length confound in the same corpus [5] and prompt-specific collection shortcuts [6]. What is missing is a way to ask how much, in total, a benchmark's separability owes to surface form rather than content, and to place two benchmarks on the same axis.

This paper proposes that measurement. We compare three arms on identical splits. A *surface-only* model reads 47 orthographic features and never sees a word. A *content-only* model reads text after punctuation, casing and non-ASCII characters are stripped. A *full* model is a fine-tuned transformer on raw text. The first two share a classifier family and are directly comparable, and the third is a reference point rather than a matched arm.

Applied to DAIGT V2 and HC3, the answer differs sharply. On HC3 the two arms are indistinguishable, 3.20% against 3.26% test error, while on DAIGT V2 content is stronger by a factor of 7.9. Splitting each corpus by its own subgroup labels then shows the HC3 tie is not a property of the benchmark. It belongs to the Reddit sub-corpus that is three quarters of it, and reverses on the other domains.

Our contributions are three. The first is the decomposition, stated formally in Section III-C as a reusable measurement over any human-versus-machine corpus, with the length control that stops its two arms sharing a channel. The second is its application to two benchmarks and their eighteen sub-corpora, which locates the HC3 result in one collection convention in one dominant domain and shows that removing that convention costs the surface arm ten points. The third is a tokenisation control establishing that the most-discussed cue in HC3 is sufficient in isolation yet unnecessary in practice, since BERT provably cannot represent it and still reaches 0.9916 weighted F1 there.

II. Related Work

Detection on these benchmarks. Guo et al. [1] introduce HC3 with a RoBERTa detector at 99.82 F1, establishing both the corpus and the accuracy regime the field has worked in since, though Su et al. [2] show that regime narrows to roughly 91.7% under semantic-invariant tasks. Zhou [3] reaches competitive DAIGT V2 accuracy with character n-grams and no transformer, a result consistent with ours but reported without the surface-content separation that would explain it. Yadagiri et al. [7] pair DeBERTa with stylometric features and Socolof et al. [8] detect with style embeddings alone, both combining the two channels to raise a score where we hold them apart to measure one.

The closest prior result. Tian et al. [4] identify the HC3 whitespace convention, release a cleaning kit, and go further than they are usually credited for. Their appendix reports that a detector consisting of a single test for one token identifier reaches 82.12 F1 on HC3 at sentence level, above the 81.89 they quote for a fine-tuned RoBERTa. Our document-level equivalent of that rule reaches 94.23, and the difference is sentence against document rather than a disagreement. Theirs is the closest existing statement of this paper's HC3 finding. What we add is a matched content arm to weigh the cue against, a per-domain split that locates it, and a tokenisation control asking whether a given detector can represent the cue at all.

Shared benchmarks and shortcut learning. RAID [9], M4 [10] and SemEval-2024 Task 8 [11] make detectors comparable across generators, domains and attacks. They do not ask what any one condition is separable by, which is the axis added here and the reason this measurement would apply to them unchanged. The pattern is not specific to text detection. Torralba and Efros [12] showed vision datasets are identifiable from their own images, and in natural language inference a model reading only the hypothesis reaches far above chance [13], [14], with Geirhos et al. [15] giving the general account. Our surface arm is a partial-input baseline transferred to this task, except that it removes a property of the text rather than a span of it. Feng et al. [16] give the caveat that governs how such a baseline may be read. A partial-input model scoring highly shows a dataset is cheatable, while one scoring poorly does not show it is clean, since the restricted representation may fail to express an artefact that is present. We therefore report the surface arm as an upper bound on surface-carried separability rather than as a measurement of it.

Robustness and fairness. Antoun et al. [17] attack HC3-trained detectors with misspellings and homoglyphs and report a fall from 99.88 F1 to 33.57% accuracy. Liang et al. [18] show deployed detectors misclassify non-native English writing as machine-generated at high rates, which bears on ours, since a detector whose separability rests on orthographic regularity should concentrate its errors on writers whose orthographic habits differ from the corpus it was fitted to.

III. Method

A. Corpora and partitioning

DAIGT V2 [19] contains 44,868 argumentative student essays, 27,371 human-written and 17,497 machine-generated by a mixture of 2023-era systems, assembled for a public detection competition. HC3 [1] contains 85,449 question-answer rows from five English sources, contrasting human answers with GPT-3.5-Turbo. The two make a useful pair because they differ in nearly every respect that matters here. DAIGT V2 has many generators, one genre and long documents, HC3 one generator, five source domains and short ones. Both were class-balanced by downsampling, giving 34,994 and 53,806 rows, which fixes a degenerate single-class predictor at 0.333 weighted F1 rather than something respectable.

HC3 carries 6,118 duplicate rows, 7.16% of the corpus, so the split is group-aware. Documents are grouped by an MD5 hash of their whitespace-normalised, lowercased text and whole groups go to one partition. The partition is 72/8/20, since a fifth goes to test first and a tenth of the remainder to validation. Measured directly, the group-aware rule leaks 0 of 10,732 HC3 test documents where a plain stratified split of the same sample leaks 570, or 5.30%. The split is built once and reused by every model and seed.

B. Models

Five families are evaluated. Three are classical, Naive Bayes, logistic regression and a linear support vector machine, each under bag-of-words and TF-IDF. Two are transformers, `bert-base-uncased` [20] and `microsoft/deberta-v3-base` [21], fine-tuned over a sixteen-run grid per dataset, eight per model, varying learning rate, batch size and weight decay, with the operating point selected on validation weighted F1. DeBERTa's SentencePiece tokeniser [22] encodes leading whitespace whereas BERT's WordPiece does not, which makes the pair a controlled contrast on exactly the cue Section IV-D examines.

C. The decomposition

Write x for a document and $y \in \{0,1\}$ for its label, with 1 denoting machine-generated. The surface map ϕ_{S} sends a document to $\mathbb{R}^{47}$, a vector of orthographic statistics covering punctuation rates, whitespace behaviour including spaces before punctuation, casing, document and sentence length, non-ASCII and digit rates. It reads no word identity. The content map ϕ_{C} lowercases, replaces every character outside the lowercase Latin alphabet and whitespace with a space, collapses runs of whitespace, then takes raw bag-of-words counts, so punctuation, casing and non-ASCII cannot enter it.

Each arm fits one logistic regression on the same training partition,

$$h_a = \arg\min_h \sum_{(x,y)\in\mathcal{D}_{\mathrm{tr}}} \ell\big(h(\phi_a(x)), y\big) + \tfrac{1}{2C}\|h\|_2^2, \quad (1)$$

for $a \in \{\mathrm{S}, \mathrm{C}\}$, with C fixed at the library default for both. Sharing the classifier family and regularisation is what makes the two arms comparable. We report the held-out error ε_a of each arm and the gap $\Delta = \varepsilon_{\mathrm{S}} - \varepsilon_{\mathrm{C}}$. A gap near zero says orthography alone reaches the accuracy of content alone. The comparison is between two specific arms rather than a partition of the total separability, so a richer featurisation would lower either side, and it is not a claim that a corpus is clean.

One channel reaches both arms, since the surface arm reads size counts directly and raw bag-of-words rows sum to document length. Length is a documented confound on both corpora [5], so we close it, dropping the five size-scaling

TABLE I

TEST ERROR FOR ALL EIGHT MODEL CONFIGURATIONS. TRANSFORMER ROWS ARE THE DEPLOYED CHECKPOINTS, NEVER A GRID MAXIMUM. BOLD MARKS THE BEST CLASSICAL AND THE BEST OVERALL ENTRY PER COLUMN.

Model	Rep.	DAIGT V2 err %	HC3 err %
Naive Bayes	BoW	4.09	12.87
Naive Bayes	TF-IDF	4.23	13.28
Logistic Reg.	BoW	1.07	**4.49**
Logistic Reg.	TF-IDF	1.43	6.35
SVM	BoW	1.30	5.25
SVM	TF-IDF	**0.90**	5.51
BERT	raw	0.84	0.84
DeBERTa	raw	**0.83**	**0.28**

surface features and normalising content rows to unit total before rescaling by the mean training document length. The rescaling matters, since rows summing to one shrink every feature and the arm would otherwise collapse for reasons of scale rather than length.

D. Statistics and controls

Transformer baselines are read from the deployed checkpoint's own record rather than from a grid maximum. Where two models are compared on the same test set the comparison is paired. Writing b and c for the discordant counts, we report the exact two-sided binomial p-value [23] together with a paired bootstrap 95% interval on the error difference over 10,000 resamples [24], resampling document indices so the pairing is preserved. A bootstrap says nothing about how the corpus was partitioned, and the central claim here is a null, so the decomposition is repeated over five group-aware partitions varying only the split seed. We run no parametric tests at $n \leq 30$.

Two controls guard the claims below. A *tokenisation* control compares token identifier sequences for minimally different strings, since whether a detector can exploit a character-level cue is a property of its tokeniser and directly testable. A *pipeline-fires* control confirms that a cleaning step reporting no change actually executed, by applying the same pipeline to the corpus where it removes something the tokeniser can represent.

IV. RESULTS

A. A classical model matches a transformer on DAIGT V2 but not on HC3

Table I reports every configuration, and the two benchmarks part company. On DAIGT V2 the best classical configuration sits 0.0007 weighted F1 below the best transformer, and on the same documents the two disagree on 99 cases split 47 to 52, so McNemar returns $p = 0.69$ and the paired interval on the error difference, $[-0.21, +0.34]$ points, contains zero. On HC3 the same comparison gives 484 discordant cases split 16 to 468, $p < 10^{-6}$, and an error difference of $+4.21$ points.

Repeating the four deployed checkpoints at seeds 123 and 456 moves them very little. DAIGT V2 BERT averages 0.9921 over a range of 0.0011 and DeBERTa 0.9930 over 0.0024, so the two overlap. HC3 BERT averages 0.9930 over 0.0036 and DeBERTa 0.9970 over 0.0005, and those ranges are disjoint. The tie and the separation both survive reseeding, in the directions the paper needs.

TABLE II

TOP, THE DECOMPOSITION ON EACH CORPUS, WITH THE LOWER ROWS CLOSING THE DOCUMENT-LENGTH CHANNEL BOTH ARMS SHARE. BOTTOM, HC3 SPLIT BY SOURCE DOMAIN, EACH REBALANCED TO PARITY, WITH THE SHARE OF DOCUMENTS CARRYING AT LEAST ONE SPACE BEFORE PUNCTUATION. DOMAINS BELOW 200 TEST ROWS ARE MARKED, SINCE A PAIRED TEST THERE FAILS FOR WANT OF POWER RATHER THAN OF AN EFFECT.

Arm	DAIGT V2 err %	HC3 err %
surface-only	7.86	**3.20**
content-only	**0.99**	3.26
full transformer	0.83	0.28
length-only (3 feats)	24.59	15.23
surface-only, no length	9.93	**3.37**
content-only, length-norm.	**0.94**	6.28

HC3 domain	n	surface err %	content err %	cue present in human	cue present in machine
reddit_eli5	6,690	**0.01**	2.66	98.9%	0.3%
finance	684	7.60	**3.07**	5.6%	0.1%
medicine	250	4.00	**0.40**	17.0%	0.1%
open_qa	198	4.04	5.05	60.7%	0.4%
wiki_csai	168	5.36	11.90	1.6%	0.2%

open_qa and wiki_csai fall below 200 test rows.

B. On HC3, orthography alone matches content alone

Table II gives the decomposition. On HC3 the two arms are statistically indistinguishable, 3.20% error from orthography against 3.26% from content, disagreeing on 625 documents split 316 to 309, so McNemar returns $p = 0.81$ and the bootstrap puts the difference at -0.07 points with interval $[-0.52, +0.40]$. Forty-seven features that never read a word do as well as a bag-of-words model over the whole vocabulary. On DAIGT V2 the same arms separate by a factor of 7.9, 7.86% against 0.99%, with 569 discordant documents split 44 to 525 at $p < 10^{-6}$.

The content arm is deliberately unfiltered. Refitting it with the stopword removal and lemmatisation Table I's classical pipeline uses reproduces that pipeline exactly, and on HC3 the filtered arm then loses to orthography by 1.30 points where the unfiltered arm ties it. We report the unfiltered arm because it is the stronger opponent, which makes the parity claim harder rather than easier, and against it the HC3 classical-to-transformer error ratio is 11.6 to one rather than the sixteen to one Table I implies.

Surface form is informative on both benchmarks, since DAIGT V2's surface arm reaches 0.9214 weighted F1. The finding is not that one corpus is clean, but that only on HC3 does surface rise to parity with content. Closing the length channel sharpens both conclusions rather than weakening them, widening DAIGT V2's content advantage from 7.9 to 10.6 times and turning HC3's parity into a 2.91-point advantage for orthography. A null on one partition is what a lucky split manufactures, so the decomposition runs again over five group-aware partitions. Content leads on DAIGT V2 on all five at $p < 10^{-6}$. On HC3 the difference reaches significance on none, with p of 0.81, 0.27, 0.34, 0.23 and 0.78, and it changes sign between partitions, which is the strong form of the null since an underpowered real difference keeps its sign.

C. The parity belongs to one sub-domain and one cue

Both corpora carry subgroup labels, and rejoining them to the existing splits recovers a source for every row. The lower half of Table II runs the same measurement per HC3 domain. On `reddit_eli5` the surface arm reaches 0.01% error, roughly one mistake in 6,690 documents. On the two other domains with enough test rows to support a claim the ordering reverses and content wins, by 4.53 points on `finance` and 3.60 on `medicine`. Since `reddit_eli5` is 74.8% of the balanced corpus, the corpus-level parity is substantially that one domain.

The mechanism is measured rather than inferred. The space-before-punctuation cue appears in 98.9% of `reddit_eli5` human documents against 0.3% of machine ones, and decays across the rest to 1.6% on `wiki_csai`. A single boolean rule, treating a document as human when the cue is present, scores 99.22% on `reddit_eli5` and 94.23% on the whole corpus at document level, against the 82.12% Tian et al. [4] report for the sentence-level equivalent. Applying their cleaning kit, and nothing else, changes 44.5% of HC3 documents and moves the surface arm from 3.20% to 13.22% error, a loss of 10.03 points with interval $[-10.69, -9.38]$, while the content arm is untouched at 3.26%. After cleaning, orthography sits 9.96 points behind content on the corpus where it had tied.

The same treatment on DAIGT V2 pairs each machine generator against the shared human pool. Content wins on every generator, but the surface arm's error spans roughly fifteenfold across them, and two pairs of the same underlying model contributed by different people differ by factors of 4 and 2.3. Surface separability there tracks collection provenance more closely than model identity, which is the conclusion the HC3 domain split reaches by another route.

D. A model that cannot represent the cue reaches 0.9916 anyway

The natural inference from the cue's dominance is that HC3-trained detectors exploit it. Tokenisation makes that testable, and the test does not support it. BERT's WordPiece tokeniser splits on punctuation irrespective of adjacent whitespace, emitting identical identifier sequences for `"the answer is simple ."` and `"the answer is simple."` in three of three pairs tested, while DeBERTa's SentencePiece distinguishes all three. BERT cannot represent the cue and still reaches 0.9916 weighted F1 on HC3. The cue is therefore sufficient in isolation, since the surface arm that reads it reaches 0.9680, and unnecessary in practice.

This is also why removing the cue changes nothing for BERT. Whitespace cleaning on HC3 yields predictions bit-identical to the uncleaned run, which alone would be indistinguishable from a cleaning step that never executed. The same pipeline on DAIGT V2, where it also removes emoji that WordPiece represents, does change BERT's predictions, so the pipeline fires and the null is specific to whitespace. We do not attribute the BERT-to-DeBERTa difference on HC3 to this cue, since the two models differ in tokeniser, architecture, pretraining corpus and parameter count. What the experiment establishes is that a null from removing a cue is uninterpretable without first checking the model could read it. Fig. 1 collects the results above.

V. Discussion and Conclusion

The decomposition separates two benchmarks that a headline accuracy figure does not, and applied per sub-corpus it separates one benchmark from itself. On HC3 orthography alone matches content alone and a paired test cannot distinguish them on any of five partitions, while on DAIGT V2 content is stronger by a factor of 7.9. Section IV-C then narrows the first of those to one collection convention in the Reddit sub-corpus that is three quarters of the corpus, and shows that removing that convention costs the surface arm ten points. A single collection convention can be cleaned. The diffuse stylistic signal DAIGT V2's surface arm reads cannot, and may not even be an artefact.

What the measurement does not license is a claim that either corpus is clean. DAIGT V2's surface arm reaches 0.9214 weighted F1, so surface form is substantially informative on both, and Feng et al. [16] give the reason a low surface score would not have settled the question either. The finding is comparative and bounded by the arms we built.

Two results here say more about method than about these corpora. A cleaning experiment reporting no change is uninterpretable without evidence the pipeline ran, and a tokeniser that cannot represent the cue being cleaned makes the null a fact about the model rather than the corpus. Both requirements cost one forward pass and neither is standard practice.

Several limits bound every number above. The transformers see at most 128 tokens, so the DAIGT V2 transformer results describe classification from an essay's opening. The four deployed checkpoints were run at three seeds and move by 0.0005 to 0.0036 weighted F1, but the grid behind them is single-seed. The content arm is bag-of-words, so word order and syntax lie outside it, and the surface set is 47 hand-built features, so ε_S is an upper bound. Five of the eighteen sub-corpora fall below 200 test rows after rebalancing and are reported as underpowered rather than as nulls. HC3 has one generator collected at one time and both corpora are English, so the orthographic conventions the surface arm reads are not language-independent.

The measurement itself is the portable part. It needs no new annotation, it fits two logistic regressions, and it runs in minutes. It yields one number per corpus, and per sub-corpus, that a headline accuracy cannot express, namely how much of that corpus a model could pass without reading the language. We would encourage reporting it alongside any new detection benchmark, in the same way a hypothesis-only baseline is now reported alongside a natural language inference dataset [14].

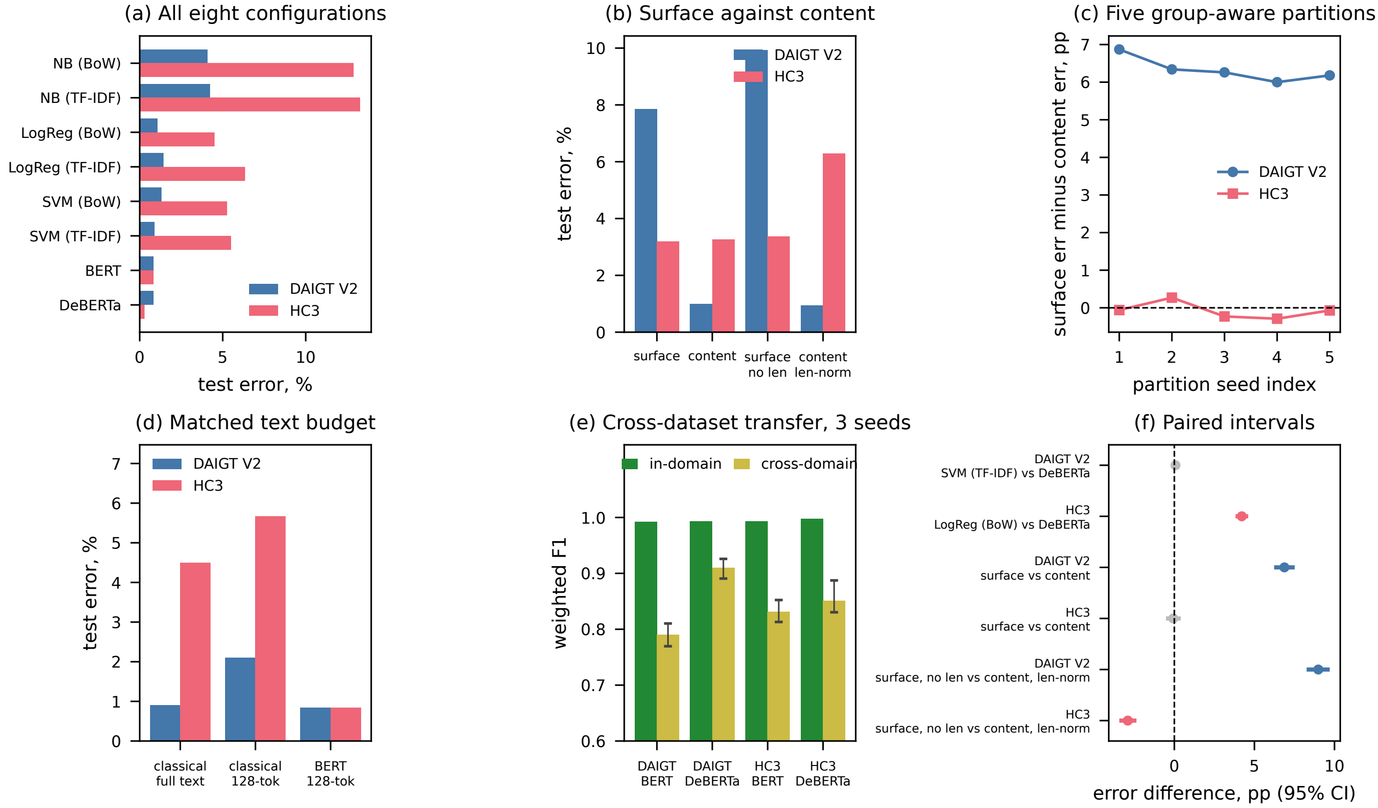

Fig. 1. Every headline result in one view. Panels give the model ranking (a), the decomposition with and without the length channel (b), the surface-minus-content difference across five partitions (c), the matched text budget (d), cross-corpus transfer (e), and the paired intervals behind each significance claim (f), where grey marks an interval containing zero.

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